Information Theory Exam, MICAS Master

Written Exam, 8 November 2023

You have 3h to solve the exam.
Try to look at all the questions in all the problems. You might find some of them easier than others. Moreover, sometimes you might be able to answer for example question 3. even if you could not answer the previous question 2. Notice also that at any given time in the exam you can use all the statements that you were asked to prove in previous parts of the exam, even if you were not able to prove them.
Make sure you justify your answers! Your answers are graded also according to your justifications.

Problem 1: Information Measures *20 points*

1. Let X be uniform over $\{1, 2, 3, 4, 5, 6\}$ and

$$\begin{aligned} Y &= X \mod 2 \\ Z &= X \mod 3, \end{aligned}$$

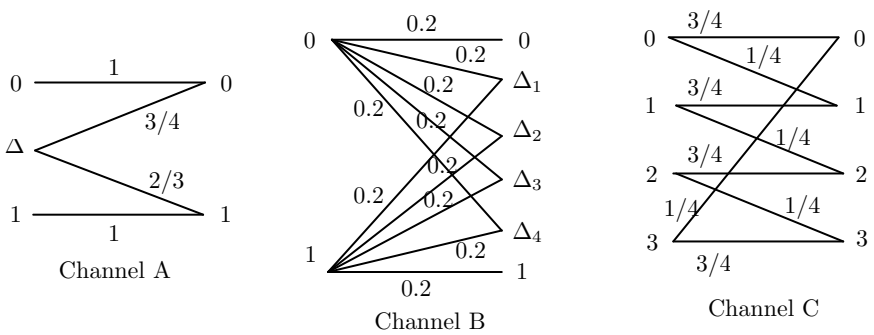
   where $\mod$ indicates the modulo operation.

   Find $H(X|Y, Z)$, $H(Y, Z)$, and $I(Y; Z)$. *(8 pts)*

2. Prove the following inequalities and find conditions for equality: *(12 pts)*

$$\begin{aligned} H(X, Y, Z) - H(X, Y) &\leq H(X, Z) - H(X) \\ I(X; Z|Y) &\geq I(Z; Y|X) - I(Z; Y) + I(X; Z). \end{aligned}$$

Problem 2: Capacity *40 points*



1. Find the capacities of channels A, B, and C in the diagramms above. *(25 pts)*

2. Fix a parameter $p \in [0, 1/2]$ and consider the two DMCs $(\mathcal{X}, \mathcal{Y}, P_{Y|X})$ and $(\mathcal{X}, \mathcal{Z}, P_{Z|X})$, where $\mathcal{X} = \mathcal{Y} = \mathcal{Z} = \{1, 2, 3, 4\}$ and

$$P_{Y|X}(y|x) = \begin{cases} 1-p & \text{if } x = y \in \{1,4\} \\ 1-2p & \text{if } x = y \in \{2,3\} \\ p & \text{if } x \in \{1,2,3\}\ y = x+1 \\ p & \text{if } x \in \{2,3,4\}\ y = x-1 \\ 0 & \text{otherwise} \end{cases}$$

and

$$P_{Z|X}(z|x) = \begin{cases} 1-p & \text{if } z = x \in \{1,2,3,4\} \\ p & \text{if } x \in \{1,3\}\ z = x+1 \\ p & \text{if } x \in \{2,4\}\ z = x-1 \\ 0 & \text{otherwise.} \end{cases}$$

Which of the two DMCs has higher capacity? Give a proof. *(15pts)*
Hint: Draw the channel diagramms for the two channels. Finding the solution to this question requires little calculations but some thinking.

Problem 3: Source Coding *30 points*

Let X be a source that takes on the values $\{1, 2, 3, 4, 5, 6, 7\}$ with probabilities $\Pr[X = k] = p_k$, where $p_1 = 1/3$, $p_2 = 1/4$, $p_3 = 1/6$, $p_4 = 1/12$, $p_5 = 1/16$, $p_6 = 1/16$, $p_7 = 1/24$.

1. Find the minimal description length for this source for the best binary prefix-free code. *(4pts)*
2. Find the minimal description length for this source for the best binary uniquely decodable code. *(4pts)*
3. Consider the following binary code: $c(1) = 10$, $c(2) = 01$, $c(3) = 11$, $c(4) = 0001$, $c(5) = 0011$, $c(6) = 0010$, $c(7) = 0000$.

   Is the code an optimal prefix-free code for our source? That means, does it attain the minimum description length for prefix-free codes? *(4pts)*
4. Is it possible that the code presented in 3. is obtained by applying Huffman's procedure to our source? Justify your answer! *(4pts)*

Assume for the rest that X is a source that takes on the values $\{1, 2, 3, 4, 5, 6, 7, 8\}$ with probabilities that satisfy $p_1 \geq p_2 \geq p_3 \geq p_4 \geq p_5 \geq p_6 \geq p_7 \geq p_8 > 0$. Assume we use a binary Huffman code to encode the source.

5. What is the smallest value p_1 such that there exists a Huffman code which assigns a codeword of length 1 to the source symbol $X = 1$? Justify your answer! *(7pts)*
6. What is the largest value for p_8 such that there exists a Huffman code which assigns a codeword of length at least 4 to the source symbol $X = 8$? Justify your answer! *(7pts)*

Problem 4: Entropy Rate *30 points*

Let $X_0, X_1, X_2, \ldots$ a sequence of independent and identically distributed Bernoulli-p random variables, for a given p in the open interval $(0, 1)$. Define the new stochastic process

$$Y_k = aX_k + X_{k-1}, \quad k = 1, 2, \ldots, \tag{1}$$

where a is a given number in $[0, 2]$. (Notice that the process $\{Y_k\}_{k=1}^{\infty}$ starts at time $k = 1$.)

1. Find the entropy rate $H(\{X_k\})$ in function of the parameter $p \in (0,1)$. *(3pts)*
2. Is the process $\{Y_k\}_{k=1}^{\infty}$ stationary? *(4pts)*
3. Is the process $\{Y_k\}_{k=1}^{\infty}$ Markov when $a = 1$? And for other values of $a \in [0,2]$? *(8pts)*
4. Find the entropy rate $H(\{Y_k\})$ in function of the coefficient a and of the parameter $p \in (0,1)$. *(15pts)*